1 Representation of $\mathfrak{sl}(2, \mathbb{C})$

1.1 The Modules V_d

1.2 Classifying the Irreducible $\mathfrak{sl}(2, \mathbb{C})$-Modules

Lemma 1.2.1

Suppose that V is an $\mathfrak{sl}(2, \mathbb{C})$-module and $v \in V$ is an eigenvector of h with eigenvalue λ.

- Either $e \cdot v = 0$ or $e \cdot v$ is an eigenvector of h with eigenvalue $\lambda + 2$.
- Either $f \cdot v = 0$ or $f \cdot v$ is an eigenvector of h with eigenvalue $\lambda - 2$.

Lemma 1.2.2

Let V be a finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$-module. Then V contains eigenvector w for h such that $e \cdot w = 0$.

Theorem 1.2.3

If V is finite-dimensional irreducible $\mathfrak{sl}(2, \mathbb{C})$-module, then V is isomorphic to one of the V_d.

Corollary 1.2.4

If V is a finite-dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$ and $w \in V$ is an h-eigenvector such that $e \cdot w = 0$, then $h \cdot w = dw$ for some non-negative interger d and the submodule of V generated by w is isomorphic to V_d.

1.3 Weyl's Theorem

Theorem 1.3.1

Let L be a complex semisimple Lie algebra. Every finite-dimensional representation of L is completely reducible.